1. ESP32-S3 MICROCONTROLLER (MASTER PINOUT MATRIX)

ESP32-S3 PIN	CONNECTED DEVICE	DEVICE PIN	SIGNAL NET	VOLTAGE LEVEL / MODE
3V3	TCA9548A, Sensors, Relay VCC	`VIN / VCC`	`+3.3V_RAIL`	3.3V DC Power (Out)
5V / VIN	3.2" SPI TFT VCC & LM2596	`VCC`	`+5V_RAIL`	5.0V DC Power (In)
GND	All Subsystems	`GND`	`SYSTEM_GND`	0V System Ground
GPIO 4	TCA9548A Multiplexer	`SDA`	`I2C_SDA`	3.3V Bi-directional (I2C Data)
GPIO 5	TCA9548A Multiplexer	`SCL`	`I2C_SCL`	3.3V Clock Out (50 kHz I2C)
GPIO 10	3.2" SPI TFT (ILI9341)	`CS`	`TFT_CS`	3.3V Digital Out (Active LOW)
GPIO 6	3.2" SPI TFT (ILI9341)	`RESET`	`TFT_RST`	3.3V Digital Out (Active LOW)
GPIO 15	3.2" SPI TFT (ILI9341)	`DC / RS`	`TFT_DC`	3.3V Digital Out (0=Cmd, 1=Data)
GPIO 17	3.2" SPI TFT (ILI9341)	`SDI (MOSI)`	`TFT_MOSI`	3.3V High-Speed SPI (40 MHz)
GPIO 18	3.2" SPI TFT (ILI9341)	`SCK (CLK)`	`TFT_SCK`	3.3V High-Speed SPI Clock
GPIO 2	3.2" SPI TFT (ILI9341)	`LED`	`TFT_LED`	3.3V PWM (Backlight Dimming)
GPIO 16	3.2" SPI TFT (ILI9341)	`SDO (MISO)`	`TFT_MISO`	3.3V Input (Optional read-back)
GPIO 9	Relay K1 (120mm Exhaust Fan)	`IN1`	`RELAY_EXH`	3.3V Digital Out (Active LOW)
GPIO 11	Relay K2 (Misting Solenoid Pump)	`IN2`	`RELAY_MIST`	3.3V Digital Out (Active LOW)
GPIO 12	Relay K3 (LED Grow Lights)	`IN3`	`RELAY_LIGHT`	3.3V Digital Out (Active LOW)
GPIO 13	Relay K4 (120mm Intake Fan)	`IN4`	`RELAY_INT`	3.3V Digital Out (Active LOW)

2. POWER DISTRIBUTION RAILS & GROUNDING

POWER RAIL	VOLTAGE	POWER SOURCE	TARGET SUBSYSTEMS & LOADS
AC Mains	220VAC	Wall Outlet → IEC C14	12V 5A Industrial Enclosed Switching PSU (5A Fused)
+12V Master Rail	+12.0V DC	12V 5A Switching PSU	Fans, Misting Pump, Grow Lights, LM2596 Buck Input
+5V Logic Rail	+5.0V DC	LM2596 Buck Converter	ESP32 5VIN pin, 3.2" TFT VCC, Relay JD-VCC coil rail
+3.3V MCU Rail	+3.3V DC	ESP32 AMS1117 LDO	ESP32 SoC, TCA9548A MUX, BH1750, SHT31, SCD41, Relay VCC

3. TCA9548A 8-CHANNEL I2C MULTIPLEXER PINOUT

TCA9548A PIN	CONNECTED TO	SIGNAL NET	HARDWARE FUNCTION & DESTINATION
VIN / VCC	ESP32 3V3 Rail	+3.3V	Powers multiplexer internal switch gates
GND	System GND	GND	Common system ground plane
SDA	ESP32 GPIO 4	I2C_SDA	Upstream master I2C data
SCL	ESP32 GPIO 5	I2C_SCL	Upstream master I2C clock (50 kHz)
RST	ESP32 3V3 Rail	+3.3V	Tied HIGH to disable hardware reset
A0, A1, A2	System GND	GND	Tied to GND → Locks I2C Address to **0x70**
SD0 / SC0	GX16 Port 1 (Pins 3, 4)	CH 0	→ 10m Cable 1 → **ROHM BH1750 Ambient Light (0x23)**
SD1 / SC1	GX16 Port 2 (Pins 3, 4)	CH 1	→ 10m Cable 2 → **Sensirion SHT31-D Temp/RH (0x44)**
SD2 / SC2	GX16 Port 3 (Pins 3, 4)	CH 2	→ 10m Cable 3 → **Sensirion SCD41 True NDIR CO2 (0x62)**
SD3–7 / SC3–7	*Leave Unconnected*	CH 3–7	Unused / Reserved for future expansion

4. GX16 4-PIN AVIATION CONNECTORS & 10-METER CABLES

GX16 PIN	WIRE COLOR	MAIN CONTROLLER SIDE	SENSOR PROBE END (CHAMBER)
Pin 1	🔴 **RED (22 AWG)**	ESP32 3V3 Rail	Sensor VCC / VIN / VDD (+3.3V)
Pin 2	⚫ **BLACK (22 AWG)**	System GND	Sensor GND (Also tie ADDR/ADR pin to GND)
Pin 3	🟢 **GREEN / BLUE**	TCA9548A SD0 / SD1 / SD2	Sensor SDA (Data Line)
Pin 4	⚪ **WHITE / YELLOW**	TCA9548A SC0 / SC1 / SC2	Sensor SCL (Clock Line)
Shell	Bare Braid Shield	System GND at Box ONLY	**LEAVE FLOATING** in chamber (Prevents ground loops)

5. SENSOR BREAKOUT BOARD PINOUTS (CHAMBER END)

SENSOR BREAKOUT	I2C ADDRESS	BREAKOUT PINS	WIRING INSTRUCTIONS
ROHM BH1750 (Lux)	0x23	VCC, GND, SDA, SCL, ADDR	VCC→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4. ADDR tied to GND.
Sensirion SHT31 (T/RH)	0x44	VIN, GND, SDA, SCL, ADR	VIN→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4. ADR tied to GND.
Sensirion SCD41 (CO2)	0x62	VDD, GND, SDA, SCL	VDD→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4.

SCD41 10-Meter Buffer Capacitor: Solder a **100µF 10V electrolytic capacitor + 0.1µF ceramic capacitor** directly across Pin 1 (VDD) and Pin 2 (GND) at the SCD41 breakout board inside the chamber probe.

6. 3.2" SPI TFT DISPLAY (ILI9341, 240X320) PINOUT

TFT PIN #	TFT PIN LABEL	ESP32-S3 PIN	SIGNAL NET	WIRING RULE & FUNCTION
1	VCC	ESP32 **5V / VIN**	+5V_RAIL	Powers display via onboard LDO (J1 MUST BE OPEN)
2	GND	ESP32 **GND**	SYSTEM_GND	Common ground plane
3	CS	ESP32 **GPIO 10**	TFT_CS	Display SPI chip select (Active LOW)
4	RESET	ESP32 **GPIO 6**	TFT_RST	Display hardware reset (Active LOW)
5	DC / RS	ESP32 **GPIO 15**	TFT_DC	Data / Command select (0=Cmd, 1=Data)
6	SDI (MOSI)	ESP32 **GPIO 17**	TFT_MOSI	SPI Master Out (40 MHz Data In)
7	SCK (CLK)	ESP32 **GPIO 18**	TFT_SCK	SPI Clock (40 MHz)
8	LED	ESP32 **GPIO 2**	TFT_LED	Backlight PWM dimming (or tie to 3.3V)
9	SDO (MISO)	ESP32 **GPIO 16**	TFT_MISO	SPI Master In (Optional read-back)
10–14	TOUCH PINS	*Optional Header*	T_CLK..IRQ	T_CLK(18), T_CS(1), T_DIN(17), T_DO(16), T_IRQ(42)
SD	SD_CS..SCK	*Unconnected*	N/C	Leave disconnected (Firebase cloud storage used)

TFT J1 SOLDER JUMPER: Copper jumper **J1** on the back of the red display PCB must remain **OPEN (UNSHORTED)** when VCC is powered from 5V. Feeding 5V with J1 shorted destroys the ILI9341 panel!

7. 4-CHANNEL 5V OPTOCOUPLED RELAY MODULE PINOUT

RELAY HEADER	CONNECTED TO	CONTROLLED DEVICE	WIRING & PROTECTION DIODE
JD-VCC	LM2596 **+5.0V** Output	Electromagnet Coil Rail	Powers relay coils (Blue jumper REMOVED)
VCC	ESP32 **3V3** Rail	Optocoupler Input LEDs	3.3V logic supply for PC817 optocouplers
GND	System **GND**	Ground Plane	Common system ground
IN1	ESP32 **GPIO 9**	Relay 1: 120mm Exhaust Fan	Active LOW. **1N4007 diode** across fan (cathode to +12V)
IN2	ESP32 **GPIO 11**	Relay 2: Misting Pump / Valve	Active LOW. **1N4007 diode** across pump (cathode to +12V)
IN3	ESP32 **GPIO 12**	Relay 3: LED Grow Light Strip	Active LOW. Resistive 12V load (no diode needed)
IN4	ESP32 **GPIO 13**	Relay 4: 120mm Intake Fan	Active LOW. **1N4007 diode** across fan (cathode to +12V)

RELAY GALVANIC ISOLATION: Pull off the blue plastic shunt connecting VCC and JD-VCC. Connect JD-VCC directly to the LM2596 +5V rail, and connect VCC to the ESP32 +3.3V pin.